

Cross-Modal Association Models (C-MAMs)

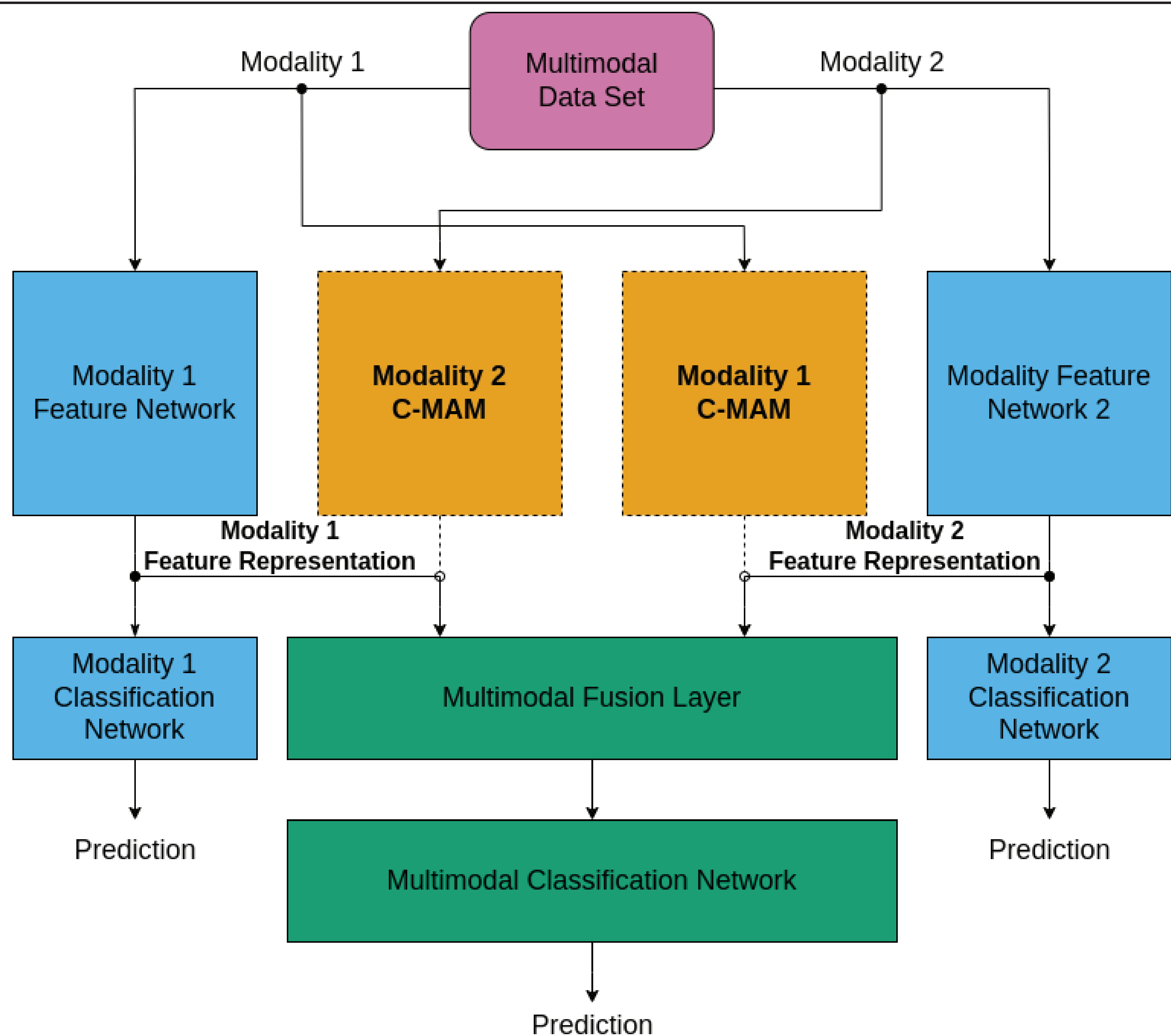
Creating Mappings Between Modality Latent Spaces to Mitigate the Impact of Missing Modality Data

Motivation

- Ever increasing amount of (fallible) data sensing and processing IoT devices which can capture multiple modalities of data.
- IoT devices can be resource constrained and geographically distributed.
- These constraints can lead to modality data going missing, for example, due to a processing delay or depleted battery.
- These devices are used in real-time applications such as:
 - Intelligent Transport Systems
 - Autonomous Vehicles
- No matter what, **a decision must be made** at regular intervals.
- How can we mitigate this risk of missing data at inference time?

Methodology

- 1) Train and evaluate expert unimodal models for the task.
- 2) Train a C-MAM for each unimodal model; Each modality is trained to predict the corresponding feature vector of the other modality.
- 3) Train and evaluate a multimodal model using your preferred training strategy, pre-trained experts or joint learning.
- 4) Evaluate the multimodal model with 100% missing modality data.
- 5) Evaluate the multimodal model with 100% missing modality data BUT using the C-MAM generated feature.



Results

MM-IMDb	Weighted	Samples	Micro	Macro
Image	43.22	44.51	45.95	27.13
Text	57.67	59.05	59.38	47.57
IT	59.63	59.02	59.09	48.48
IT w/o Image	33.34	30.94	30.50	22.90
IT C-MAM Image	54.67	58.29	58.32	38.15
IT w/o Text	14.32	23.38	23.39	11.07
IT C-MAM Text	36.78	44.11	44.43	19.46

Kinetics-Sounds	Top 1% Accuracy	Top 5% Accuracy	Precision	Recall	F1-Score
Audio	27.74	66.02	36.34	27.74	23.04
Video	77.54	94.96	79.02	77.54	77.11
AV	78.69	94.66	78.29	78.69	80.4
AV w/o Audio	77.5	93.96	78.74	77.5	76.96
AV C-MAM Audio	77.5	94.01	77.5	77.08	79.1
AV w/o Video	8.33	26.3	7.45	91.67	8.33
AV C-MAM Video	33.73	63.12	33.73	29.43	32.99

- Recovered up to 36.82% on different metrics.
- Models react differently depending on the missing modality.
- Occurs when regardless of using pre-trained unimodal models or joint learning training strategies.

Future Works

- Training with (severely) missing data and C-MAMs.
 - Pre-trained C-MAMs
 - Online training of C-MAMs
- Extending to large base models & fine-tuning per task.
- Contributing factors to C-MAM performance:
 - Network architecture
 - Feature layer selection